

# BAYAN ALAJOURE

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## SUMMARY

Final-year Electrical Engineering student at Al-Hussein Technical University (HTU) with hands-on experience in PCB design, hardware validation, and electrical system integration for power and embedded applications. Skilled in schematic capture, layout, and fabrication coordination using KiCad and EasyEDA, with a track record of taking designs from concept through functional, field-tested hardware. Seeking an Electrical Engineer / PCB Designer role.

## EDUCATION

**B.Sc. in Electrical Engineering** | Al-Hussein Technical University (HTU)

2022 – Expected 2026/2027

Focus: Electrical Systems, Control Engineering, Robotics Applications

GPA: 3.7 / 4

## WORK EXPERIENCE

**Electrical Engineer** | Albayraq Project

2024 – 2026

- Designed and developed PCBs for power circuits using KiCad, covering schematic capture, layout, fabrication coordination, and validation.
- Built a complete electrical system for an industrial curing oven used in composite manufacturing, including power distribution and wiring.
- Oversaw full electrical system integration for a 3.2 m wingspan UAV: wiring harness design, PCB assembly, power modules, and system-level testing.
- Produced and maintained technical documentation and progress reports for all electrical subsystems, supporting cross-team hardware handoffs.

**Robotics & Drone Tutor** | Code Circle at IEC Schools

2025

- Delivered a 5-week drone-building programme, covering theory, aerodynamics basics, wiring diagrams, and step-by-step build guide. 20 students assembled and flew their own UAVs by programme completion.

**Technical Assistant** | HTU Hackerspace

2024

- Supported 100+ students in designing and building electronic and mechanical prototypes from concept through functional model.
- Designed and implemented HTU Hackerspace's first borrowing and inventory management system, now used daily by 80+ students.
- Operated and maintained fabrication equipment: laser cutters, 3D printers (Prusa, Ultimaker, Ender), and CNC routers.
- Organized the National SUMO Robot Contest for three consecutive years (2024–2026), the largest SUMO contest in the Middle East, coordinating 200+ teams and 1,000+ attendees across a 3-day event.

**Accounting Manager** | Freelance

2020 – 2021

- Designed and built from scratch a fully functional Excel-based accounting and membership management system, still actively used to serve 200+ members monthly.
- Managed financial records, membership tracking, and monthly reporting for center operations.

## LEADERSHIP EXPERIENCE

**Founder & Head of Technical** | HTU Robotics Alliance (University Club)

2022 – 2023

- Founded and led the HTU Robotics Alliance, directing technical strategy and growing it into an active engineering club.
- Designed and hosted HTU's first Line Follower Robot Competition: built the track, developed the ruleset, and trained 50+ participants from 16 teams.
- Conducted 5+ technical training sessions mentoring 20+ students in robotics, microcontroller programming, and embedded systems.

## PROJECTS

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### Bunyan — Modular Robot | Capstone Project | 1st Place, HTU Capstone Expo

2026

Bunyan is a modular robotics system, consisting of a builder robot and passive voxel cells. The builder autonomously picks up voxels and places them in precise positions to construct structures, with the long-term vision of robots building other robots.

#### My contributions (Phase 1):

- Developed a fully verified mathematical model for one robot limb, covering complete kinematic and dynamic analysis, tested and validated in Autodesk Inventor
- Built a digital twin from scratch integrating MATLAB App Designer and Autodesk Inventor: users define a part, then control its parameters in real time via MATLAB sliders with live updates reflected instantly in Inventor
- Prototyped components using SLA resin, PETG, and PLA 3D printing

### FPGA Development | Intel DE1-SoC | University Coursework

2024 – 2025

- Digital Design Course: Programmed an 8-bit memory system in Verilog on the Intel DE1-SoC FPGA; covered hardware description, SD card flashing, and terminal-based development workflow.
- Embedded Systems Course: Integrated an ultrasonic distance sensor, LCD display, and speaker on the DE1-SoC: The system outputs a numeric distance reading and generates a tone triggered by either measured distance or clap sound detection.

## ACHIEVEMENTS & AWARDS

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- 1st Place — Capstone Expo | Al-Hussein Technical University | June 2026
- 2nd Place — Robofight Competition | Applied Private University, Jordan | April 2026
- 1st Place — Capstone Expo | Albayraq UAV | Al-Hussein Technical University | June 2025
- 2nd Place — Capstone Expo | Albayraq UAV | Al-Hussein Technical University | January 2026

## CERTIFICATES & COURSES

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- SOLIDWORKS Additive Manufacturing Associate (CSWA-AM)

## SKILLS & LANGUAGES

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**Hardware & PCB:** PCB Design & Fabrication • Schematic Capture & Layout (KiCad, EasyEDA, Altium Designer) • Power Modules & Distribution • Sensor Systems (IMU, GPS, Barometer, Ultrasonic) • Microcontrollers & Embedded Boards • FPGA (Intel DE1-SoC) • Communication Protocols (I2C, SPI, UART, Wireless)

**Software & Simulation:** MATLAB & Simulink • SolidWorks (CSWA) • KiCad • EasyEDA • Multisim • BetaFlight

**Lab & Fabrication:** Soldering • Oscilloscope • Multimeter • 3D Printing • Laser Cutting • CNC Routing

**Professional:** Technical Documentation • Cross-functional Coordination • Technical Leadership • Mentoring & Teaching

**Languages:** Arabic (Native) • English (C1 – Professional Proficiency) • French (A1 – Beginner)